Skills Review

Exercise No. 11

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Course/Section: CPE4B **Date: Oct. 7, 2026**

Machine Problem:

Create a program that will display the color-coding values of the resistor using Interrupt Function 10h. Use string character printing and cursor positioning to display whatever information needs to be displayed in each Resistor color. Graph the coordinates by drawing the expected output. Save the program using Exer11.asm as the filename.

Program:

```
; AARON LUDWIG A. ALTAR
; CPE4B
; Exer11 - Resistor Color Code (Interrupt 10h)

.model small
.stack
.data

var1    DB 0DBh, 0DBh, 0DBh, 0DBh, 0DBh, 0DBh, 0DBh, 0DBh, 0DBh, 0DBh,
0DBh, 0DBh,'$'
var2    DB 0B1h, 0B1h, 0B1h, 0B1h, 0B1h, 0B1h, 0B1h, 0B1h, 0B1h, 0B1h,
0B1h, 0B1h,'$'
var3    DB 'RESISTOR COLOR CODE CHART','$'
var4    DB '5    6    x100      2%','$'
var5    DB 'Green-Blue-Red-Red = 56 x 100 = 5,600 Ohms (5.6 kOhms)
+/-2%','$'
var6    DB 'BAND','$'
var7    DB 'COLOR     DIGIT   MULTIPLIER   TOLERANCE','$'
var8    DB 'BLACK     0       x1           -','$'
var9    DB 'BROWN     1       x10          +/-1%','$'
var10   DB 'RED       2       x100         +/-2%','$'
var11   DB 'ORANGE    3       x1K          -','$'
var12   DB 'YELLOW    4       x10K         -','$'
var13   DB 'GREEN     5       x100K        +/-0.5%','$'
var14   DB 'BLUE      6       x1M          +/-0.25%','$'
var15   DB 'VIOLET    7       x10M         +/-0.1%','$'
var16   DB 'GRAY      8       x100M        +/-0.05%','$'
var17   DB 'WHITE     9       x1G          -','$'
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```
var18   DB 'GOLD      -       x0.1         +/-5%','$'
var19   DB 'SILVER    -       x0.01        +/-10%','$'
var20   DB 'NONE      -       -            +/-20%','$'

.code
start:

        ; screen background (0,0 - 24,79)
        mov ah, 06h             ; Function 06h - scroll up window
        mov bh, 0Fh             ; black background, white text
        mov ch, 0               ; start row
        mov cl, 0               ; start column
        mov dh, 24              ; end row
        mov dl, 79              ; end column
        int 10h                 ; BIOS video interrupt - display the window

        ; title bar (0,0 - 0,79)
        mov ah, 06h
        mov bh, 1Eh             ; blue background, yellow text
        mov ch, 0
        mov cl, 0
        mov dh, 0
        mov dl, 79
        int 10h

        ; <---- resistor drawing ---->
        ; lead wire (4,8 - 4,71)
        mov ah, 06h
        mov bh, 70h             ; white (silver)
        mov ch, 4
        mov cl, 8
        mov dh, 4
        mov dl, 71
        int 10h

        ; resistor body (2,24 - 6,55)
        mov ah, 06h
        mov bh, 30h             ; cyan
        mov ch, 2
        mov cl, 24
```

```
        mov dh, 6
        mov dl, 55
        int 10h

        ; 1st band - green (5) (2,28 - 6,29)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 2
        mov cl, 28
        mov dh, 6
        mov dl, 29
        int 10h

        ; 2nd band - blue (6) (2,33 - 6,34)
        mov ah, 06h
        mov bh, 10h             ; blue
        mov ch, 2
        mov cl, 33
        mov dh, 6
        mov dl, 34
        int 10h

        ; 3rd band - red (x100) (2,38 - 6,39)
        mov ah, 06h
        mov bh, 40h             ; red
        mov ch, 2
        mov cl, 38
        mov dh, 6
        mov dl, 39
        int 10h

        ; 4th band - red (+/-2%) (2,48 - 6,49)
        mov ah, 06h
        mov bh, 40h             ; red
        mov ch, 2
        mov cl, 48
        mov dh, 6
        mov dl, 49
        int 10h
```

```
        ; <---- color code chart ---->
        ; chart panel (10,6 - 23,73)
        mov ah, 06h
        mov bh, 30h             ; cyan, black text
        mov ch, 10
        mov cl, 6
        mov dh, 23
        mov dl, 73
        int 10h

        ; chart header (10,6 - 10,73)
        mov ah, 06h
        mov bh, 1Eh             ; blue, yellow text
        mov ch, 10
        mov cl, 6
        mov dh, 10
        mov dl, 73
        int 10h

        ; black swatch (11,8 - 11,19)
        mov ah, 06h
        mov bh, 00h             ; black
        mov ch, 11
        mov cl, 8
        mov dh, 11
        mov dl, 19
        int 10h

        ; brown swatch (12,8 - 12,19)
        mov ah, 06h
        mov bh, 60h             ; brown
        mov ch, 12
        mov cl, 8
        mov dh, 12
        mov dl, 19
        int 10h

        ; red swatch (13,8 - 13,19)
        mov ah, 06h
        mov bh, 40h             ; red
```

```
        mov ch, 13
        mov cl, 8
        mov dh, 13
        mov dl, 19
        int 10h

        ; orange swatch (14,8 - 14,19)
        mov ah, 06h
        mov bh, 4Eh             ; red background, yellow text
        mov ch, 14
        mov cl, 8
        mov dh, 14
        mov dl, 19
        int 10h

        ; yellow swatch (15,8 - 15,19)
        mov ah, 06h
        mov bh, 0Eh             ; black background, yellow text
        mov ch, 15
        mov cl, 8
        mov dh, 15
        mov dl, 19
        int 10h

        ; green swatch (16,8 - 16,19)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 16
        mov cl, 8
        mov dh, 16
        mov dl, 19
        int 10h

        ; blue swatch (17,8 - 17,19)
        mov ah, 06h
        mov bh, 10h             ; blue
        mov ch, 17
        mov cl, 8
        mov dh, 17
        mov dl, 19
```

```
        int 10h

        ; violet swatch (18,8 - 18,19)
        mov ah, 06h
        mov bh, 50h             ; magenta (violet)
        mov ch, 18
        mov cl, 8
        mov dh, 18
        mov dl, 19
        int 10h

        ; gray swatch (19,8 - 19,19)
        mov ah, 06h
        mov bh, 08h             ; black background, gray text
        mov ch, 19
        mov cl, 8
        mov dh, 19
        mov dl, 19
        int 10h

        ; white swatch (20,8 - 20,19)
        mov ah, 06h
        mov bh, 0Fh             ; black background, white text
        mov ch, 20
        mov cl, 8
        mov dh, 20
        mov dl, 19
        int 10h

        ; gold swatch (21,8 - 21,19)
        mov ah, 06h
        mov bh, 6Eh             ; brown background, yellow text
        mov ch, 21
        mov cl, 8
        mov dh, 21
        mov dl, 19
        int 10h

        ; silver swatch (22,8 - 22,19)
        mov ah, 06h
```

```
        mov bh, 70h             ; white (silver)
        mov ch, 22
        mov cl, 8
        mov dh, 22
        mov dl, 19
        int 10h

        mov ax, @data           ; initialize the data segment
        mov ds, ax

        ; <---- title and resistor values ---->
        mov ah, 02h             ; Function 02h - set cursor position
        mov bh, 00h             ; page 0
        mov dh, 0               ; row
        mov dl, 27              ; column
        int 10h
        mov ah, 09h             ; Function 09h - display string
        mov dx, offset var3     ; "RESISTOR COLOR CODE CHART"
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 7
        mov dl, 28
        int 10h
        mov ah, 09h
        mov dx, offset var4     ; value under each band
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 8
        mov dl, 10
        int 10h
        mov ah, 09h
        mov dx, offset var5     ; sample reading of the resistor
        int 21h

        ; <---- fill the swatches that need a brighter / mixed color ---->
```

```
        ; 0DBh = full block (solid text color), 0B1h = shaded block (mix
of text and background)
        mov ah, 02h
        mov bh, 00h
        mov dh, 14
        mov dl, 8
        int 10h
        mov ah, 09h
        mov dx, offset var2    ; orange swatch - shaded blocks
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 15
        mov dl, 8
        int 10h
        mov ah, 09h
        mov dx, offset var1    ; yellow swatch - full blocks
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 19
        mov dl, 8
        int 10h
        mov ah, 09h
        mov dx, offset var1    ; gray swatch - full blocks
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 20
        mov dl, 8
        int 10h
        mov ah, 09h
        mov dx, offset var1    ; white swatch - full blocks
        int 21h

        mov ah, 02h
        mov bh, 00h
```

```
        mov dh, 21
        mov dl, 8
        int 10h
        mov ah, 09h
        mov dx, offset var2    ; gold swatch - shaded blocks
        int 21h

        ; <---- chart text ---->
        mov ah, 02h
        mov bh, 00h
        mov dh, 10
        mov dl, 8
        int 10h
        mov ah, 09h
        mov dx, offset var6    ; header - swatch column
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 10
        mov dl, 22
        int 10h
        mov ah, 09h
        mov dx, offset var7    ; header - value columns
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 11
        mov dl, 22
        int 10h
        mov ah, 09h
        mov dx, offset var8    ; black values
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 12
        mov dl, 22
        int 10h
```

```
        mov ah, 09h
        mov dx, offset var9    ; brown values
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 13
        mov dl, 22
        int 10h
        mov ah, 09h
        mov dx, offset var10   ; red values
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 14
        mov dl, 22
        int 10h
        mov ah, 09h
        mov dx, offset var11   ; orange values
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 15
        mov dl, 22
        int 10h
        mov ah, 09h
        mov dx, offset var12   ; yellow values
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 16
        mov dl, 22
        int 10h
        mov ah, 09h
        mov dx, offset var13   ; green values
        int 21h
```

```
        mov ah, 02h
        mov bh, 00h
        mov dh, 17
        mov dl, 22
        int 10h
        mov ah, 09h
        mov dx, offset var14   ; blue values
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 18
        mov dl, 22
        int 10h
        mov ah, 09h
        mov dx, offset var15   ; violet values
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 19
        mov dl, 22
        int 10h
        mov ah, 09h
        mov dx, offset var16   ; gray values
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 20
        mov dl, 22
        int 10h
        mov ah, 09h
        mov dx, offset var17   ; white values
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 21
        mov dl, 22
```

```
        int 10h
        mov ah, 09h
        mov dx, offset var18   ; gold values
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 22
        mov dl, 22
        int 10h
        mov ah, 09h
        mov dx, offset var19   ; silver values
        int 21h

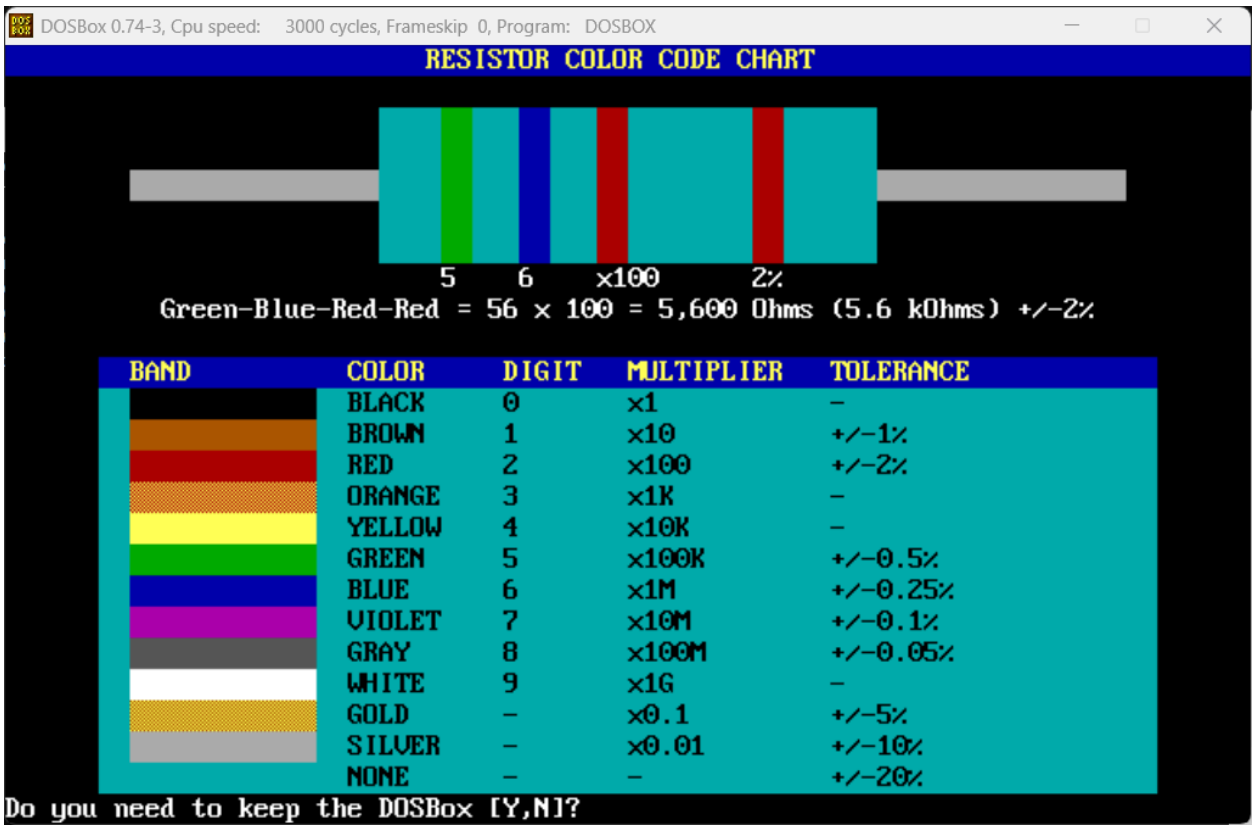
        mov ah, 02h
        mov bh, 00h
        mov dh, 23
        mov dl, 22
        int 10h
        mov ah, 09h
        mov dx, offset var20   ; no 4th band
        int 21h

        mov ah, 02h            ; move the cursor near the bottom so the
        mov bh, 00h            ; DOS prompt will not cover the design
        mov dh, 23
        mov dl, 0
        int 10h

        mov ah, 4Ch            ; terminate program
        int 21h

end start
```

Output:


DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DOSBOX
RESISTOR COLOR CODE CHART
5 6 x100 2%
Green-Blue-Red-Red = 56 x 100 = 5,600 Ohms (5.6 kOhms) +/-2%
BAND COLOR DIGIT MULTIPLIER TOLERANCE
BLACK 0 x1 -
BROWN 1 x10 +/-1%
RED 2 x100 +/-2%
ORANGE 3 x1K -
YELLOW 4 x10K -
GREEN 5 x100K +/-0.5%
BLUE 6 x1M +/-0.25%
VIOLET 7 x10M +/-0.1%
GRAY 8 x100M +/-0.05%
WHITE 9 x1G -
GOLD - x0.1 +/-5%
SILVER - x0.01 +/-10%
NONE - - +/-20%
Do you need to keep the DOSBox [Y,N]?